



Practice article

Model-free adaptive filter to mitigate actuator wear

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ABSTRACT

The present article introduces an adaptive filter of statistical basis developed for closed-loop control applications, whose goal is to reduce actuator wear while ensuring a similar control performance regarding the original closed-loop system. The main idea is to avoid the rapid change of the filtered signal when the system output has a derivative not statistically significant regarding the expected measurement noise. The adaptation law of the time constant of the filter is model-free, and the only required information is the variance of the additive noise that the measurements are subjected to. The performance of the proposed adaptive method is illustrated through a combined numerical and experimental study, in addition to its application in an operational oil plant. The results indicate that our formulation holds promise for extending the life of actuators and is easy to implement in most programmable logic controllers.

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1. Introduction

Maintenance and failure costs for industrial plants can negatively affect the business' profitability, reaching high values depending on the application type. Within this context, an essential aspect of industrial control is the trade-off between load rejection and noise attenuation. Controllers with high gains can improve the output response to load disturbance, but the resulting overuse of actuators is responsible for failures and operational interruptions [1]. Filtering the measurement signal can alleviate the effects of measurement noise, but heavy filtering yields lag in the control loop and degrades the overall performance.

In industry, regulatory control requires only a few easy-to-tune intuitive parameters affecting the control loop. For this reason, many engineers prefer to avoid the derivative action, which increases the noise sensitivity, or to detune the control gain, to design an extra filter component. For instance, a detuning technique is proposed in [2,3] to obtain a smoother control action

with less sensitivity to measurement noise. However, the use of conservative gains has major impact in the loop performance.

Designing an appropriate filter for the measurement signal that balances robustness, load disturbance attenuation and measurement noise injection has attracted the attention of the control community in the last decades [4–12]. Methods in [4–6,9] are based on optimization problems, and in [7,8], a filter design procedure is presented for a first-order plus time-delay approximation, where the proportional–integral–derivative (PID) control must be tuned iteratively for some given parameter. In search for simplicity, the filter time constant can be chosen by empirical rules [4]. In [9], the controller and filter parameters are simultaneously optimized, as opposed to the standard sequential design. An algorithm on discrete-time filtered PID controller design with thirteen sequential steps is presented in [10]. A noise observer is proposed in [11,12] by adding an inner loop in the conventional feedback loop. The design of the filter in [11] is simple, but the tracking ability is preserved only if the internal model is accurate enough. On the other hand, the observer structure presented in [12] explicitly depends on the controller and plant model. Thus, the main difference between the approaches for noise observations is whether the observer is controller-dependent, as in [12], or not, as in [11]. Finally, Kalman filters can provide optimal state estimation that delivers higher noise rejection performance than filters based on the frequency domain approach if both model and observation errors are zero-mean and uncorrelated in time [13].

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However, for an accurate estimation the linear dynamics and the process and measurement noise covariances must be known.

The works mentioned above have two main drawbacks: the filter parameters are static and the controller gains must be designed (or re-designed) simultaneously or in an iterative procedure. Furthermore, the filter design is model-dependent, and the performance of the measurement noise rejection depends on the accuracy level of system model. The design of an ideal controller with the later addition of a filter is considered ineffective if a fixed time constant for the filter is considered [7]. Finally, implementing filters in large industrial plants with control loops that are already operational is prohibitive if all the loops must be redesigned or if precise models are required. This motivates us to seek model-free adaptive filter design techniques.

Inspired by the behavior of a modified Kalman filter presented in [14], an adaptive filter is proposed based on the statistical behavior of the measured signal and filter output. Its main motivation is to be filtering design technique with low computational costs for large-scale implementation where precise models are not readily available. This approach allows the filter to suppress additive measure noise during stationary state while maintaining its reactivity to significant changes in the input signal. As a requirement, the standard deviation or variance of the measurement noise is previously defined and used as a reference for calculating the probability of the amplitude difference between samples being driven by noise or by changes in the input signal. Once this analysis has been made, a weighted sum of the measured sample and the last filter output is adapted to change the filter reactivity and produce the next output.

A practical usage of this adaptive filter is presented in PID control loops, which are by far the most used control strategies in the industry. Usual industrial processes stay close to a fixed set-point most of the time and are subjected to smooth disturbances. During this time, the PID continuously changes the manipulated variable (system input) mostly because of measurement noise and system disturbances commonly associated with the derivative part of the controller. Whenever these changes happen, the controller may result in faster wearing of the actuators, therefore raising maintenance costs. These costs can scale up to expensive and significant values in industries with heavy actuator usage, for instance the oil and gas industry [15]. The adaptive filter can be directly inserted in a PID control loop with additive white Gaussian noise (AWGN) characteristics to reduce variations in the control signal without compromising the performance of the controller on the process variable (PV). Numerical examples are shown to support the efficiency of the filter in reducing the impact of noise while maintaining the control loop performance.

The experimental validation of the presented filter has been performed on a liquid level two-tank system. After validation, the filter has been tested on the supervisory system of a Brazilian oil refinery, providing real-life industrial results of its application. The algorithm is designed to be implemented on most programmable logical controllers (PLC), thus it offers more than one way to add into a previously designed control loop. For each controller, the system response using the adaptive filter was compared with the unfiltered PID response and quantified by performance indexes such as total variation (TV), integrated absolute error (IAE) and integrated squared error (ISE).

Compared with the existing results, the paper has the following main contributions:

- A novel concept of statistical-based filter for control loops.
- Filter with negligible impact on system performance while significantly reducing actuator usage.
- Model-free design since the variance of measurement device is the only required design parameter.

- Easy-to-use solution for large-scale implementation verified in an industrial oil plant.

Our paper is organized as follows. Section 2 explains the design procedure and mathematical description of the adaptive filter, which is capable of reducing noise while maintaining step response. In Section 3 we present numerical results and experimental tests of the adaptive filter in a coupled tank plant. After validation, Section 4 shows the results obtained from the real-world industrial application in an oil refinery and the tuning of the adaptive filter for different cases. The conclusions and suggestions for further research are given in the last section.

2. Adaptive filter

For the description of the proposed adaptive filter, we first define a generic feedback control system for an industrial plant shown in Fig. 1, where u and y represent the control input and the process output, respectively. The signal n is an additive measurement noise assumed to be a random variable with the following time-invariant statistical properties: zero-mean and standard deviation σ_n . Thus, the measured output is defined as $y_l = y + n$.

The basic functionality of the proposed adaptive filter (see Fig. 2) is to analyze how the last sample of the filter output and the current sample of the measured variable are statistically different according to σ_n . If they are, then the filter detects the current sample of the measured output as a relevant modification of the process output and propagates the corresponding changes to the controller. Otherwise, the filter tries to minimize the disturbances that arrive into the controller.

As a consequence, the proposed adaptive filter focuses on reducing the impact of noise propagation on the control signal due to minor modifications to the process output without bringing significant changes to the original closed-loop performance (i.e., without the adaptive filter). It is also important to note that the filter in question can be used regardless of controller type or plant model.

To express the dynamic behavior of the proposed adaptive filter through a suitable mathematical tool, we propose a moving average approach. In spite of its simplicity, such approach reduces random noise while retaining a sharp step response. This equation has been well-understood for many decades and extensively used in statistical analysis [16].

Among a variety of different moving average methods that can be found in the literature, we choose the exponential moving average (EMA) [17] as the format of the proposed adaptive filter because it is computationally efficient and it is more responsive to new information for the same level of white noise attenuation when compared to the simple moving average.

By defining y_l as the filter input (i.e., measured output) and y_o as the filter output, the exponential moving average can be calculated recursively as follows:

$$\begin{cases} y_o[0] = y_l[0] \\ y_o[k] = y_o[k-1] + \alpha[k](y_l[k] - y_o[k-1]), \quad k > 0 \end{cases} \quad (1)$$

where $0 \leq \alpha \leq 1$ is a weight coefficient, whose value determines the smoothing of y_l . Note that the EMA can be interpreted as a first-order low pass filter in its usual computational implementation, where the parameter α actually reflects its time constant. It remains to define an expression to obtain α .

Regarding the dynamic behavior of the proposed adaptive filter, we aim to use α to represent how statistically confident we are that the filter output reflects the same direction of the process output at each sampled time k . Thus we define this statistical confidence as the difference of the probability that the filter

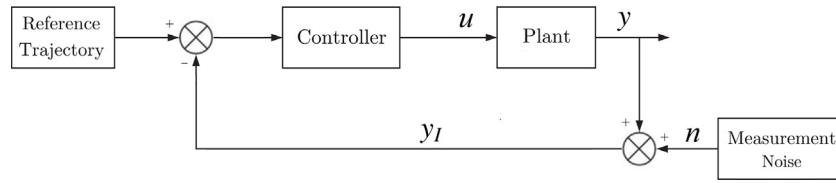


Fig. 1. Block diagram of measurement noise propagation in the closed-loop control.

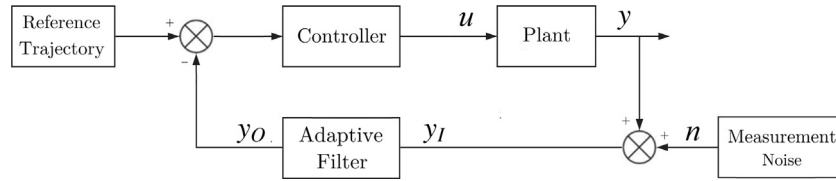


Fig. 2. Addition of the adaptive filter to the closed-loop control.

update $y_I[k] - y_O[k - 1]$ is acting so that y_O approaches y , minus the probability that it is not.

For example, in case $y_I[k]$ is just slightly larger than $y_O[k - 1]$, both probabilities are close to 50% since one cannot guarantee that y is increasing or that this difference is only due to the measurement noise, so $y_I[k]$ will not have a significant impact on $y_O[k]$ (i.e., $\alpha \approx 0$). On the other hand, in the case the measured output is considerably different from the filter output according to σ_n , the filter has no effect (i.e., $\alpha = 1$), making the closed-loop response for large disturbances identical with the one without the filter.

Within this context, let Y_O and Y_I be random processes that represent the filter input and output, respectively. Then, we propose to adjust α at each sampled time k through the following probability equation:

$$\alpha[k] = P(\text{sign}(Y_O[k - 1] - Y_I[k]) = \text{sign}(\mu_{Y_O[k-1]} - \mu_{Y_I[k]})) - P(\text{sign}(Y_O[k - 1] - Y_I[k]) \neq \text{sign}(\mu_{Y_O[k-1]} - \mu_{Y_I[k]})) \quad (2)$$

where $Y_O[k - 1]$ and $Y_I[k]$ are observations of Y_O with mean $\mu_{Y_O[k-1]}$ and Y_I with mean $\mu_{Y_I[k]}$, respectively. Here $\text{sign}(x)$ represents the sign function, which is defined as:

$$\text{sign}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases} \quad (3)$$

Eq. (2) can be rewritten as:

$$\alpha[k] = 2P(\text{sign}(Y_O[k - 1] - Y_I[k]) = \text{sign}(\mu_{Y_O[k-1]} - \mu_{Y_I[k]})) - 1 \quad (4)$$

which allows us to analyze the following cases from (3):

$$(i) \mu_{Y_O[k-1]} - \mu_{Y_I[k]} > 0$$

$$\begin{aligned} \alpha[k] &= 2P(\text{sign}(Y_O[k - 1] - Y_I[k]) = 1) - 1 \\ &= 2P(Y_O[k - 1] - Y_I[k] > 0) - 1 \end{aligned} \quad (5)$$

$$(ii) \mu_{Y_O[k-1]} - \mu_{Y_I[k]} = 0$$

$$\begin{aligned} \alpha[k] &= 2P(\text{sign}(Y_O[k - 1] - Y_I[k]) = 0) - 1 \\ &= 2P(Y_O[k - 1] - Y_I[k] = 0) - 1 \end{aligned} \quad (6)$$

$$(iii) \mu_{Y_O[k-1]} - \mu_{Y_I[k]} < 0$$

$$\begin{aligned} \alpha[k] &= 2P(\text{sign}(Y_O[k - 1] - Y_I[k]) = -1) - 1 \\ &= 2P(Y_O[k - 1] - Y_I[k] < 0) - 1 \\ &= 2P(-(Y_O[k - 1] - Y_I[k]) > 0) - 1 \end{aligned} \quad (7)$$

From the above analysis, it follows that if we define a random process Y_{OI} , in which an observation at instant k has mean and variance given by

$$\mu_{Y_{OI}[k]} = |\mu_{Y_O[k-1]} - \mu_{Y_I[k]}| \quad (8)$$

$$\sigma_{Y_{OI}[k]}^2 = \sigma_{Y_O[k-1]}^2 + \sigma_{Y_I[k]}^2 + 2Cov(Y_O[k - 1], Y_I[k]) \quad (9)$$

where $\sigma_{Y_O[k-1]}^2$ and $\sigma_{Y_I[k]}^2$ are variances of $Y_O[k - 1]$ and $Y_I[k]$, respectively, Eq. (4) can then be rewritten as follows:

$$\alpha[k] = 2P(Y_{OI}[k] \geq 0) - 1 \quad (10)$$

Eq. (10) provides the theoretical value of the weight coefficient of (1) that reflects how statistically confident we are that the filter output reflects the same direction of the process output at each sampled time k . In order to ease its implementation in practical use, now we investigate (10) under the following technical assumptions:

Assumption 1. The random variable $Y_{OI}[k]$ can be approximated by a normal distribution with mean $\mu_{Y_{OI}[k]}$ and variance $\sigma_{Y_{OI}[k]}^2$.

Remark 1. In practical applications, a reasonable assumption for the source of the sensor measurement noise is the composition of statistically independent elements such as sensor resolution, unknown environment disturbances, and nonlinear sensor dynamics. Within this context, from the central limit theorem, the total contribution of these elements tends toward a normal distribution even if the original variables themselves are not normally distributed. Therefore, we assume that the statistical properties of the sensor measurement noise can be approximated to a normal distribution, which implies the same for the observations from both the filter's input and output. Consequently, Y_{OI} can also be approximated by a normal distribution since it is defined based on a composition of these observations.

Assumption 2. Both high frequency and random behavior of the measurement noise implies that the observations $Y_O[k - 1]$ and $Y_I[k]$ can be considered approximately statistically independent of each other. Mathematically, it implies that $Cov(Y_O[k - 1], Y_I[k]) \sim 0$.

Assumption 3. The variance of the filter output is usually significantly smaller than the variance of the measured output.

Assumption 4. By considering an additive zero-mean measurement noise, the samples $y_O[k - 1]$ and $y_I[k]$ can be used to estimate $\mu_{Y_O[k-1]}$ and $\mu_{Y_I[k]}$, respectively.

From [Assumption 1](#), Eq. (10) can be rewritten as:

$$\begin{aligned}\alpha[k] &= 2P(\mathcal{N}(\mu_{Y_{OI}}, \sigma_{Y_{OI}}^2) > 0) - 1 \\ &= 2P(\mathcal{N}(0, \sigma_{Y_{OI}}^2) \leq \mu_{Y_{OI}}) - 1\end{aligned}\quad (11)$$

which can then be expressed in terms of the Gauss error function as follows:

$$\alpha[k] = \operatorname{erf}\left(\frac{|\mu_{Y_{OI}}[k-1] - \mu_{Y_I}[k]|}{\sqrt{2(\sigma_{Y_{OI}}^2[k-1] + \sigma_{Y_I}^2[k] + 2\operatorname{Cov}(Y_{OI}[k-1], Y_I[k])}}}\right)\quad (12)$$

Next, from [Assumptions 2–4](#), it follows that Eq. (12) can then be simplified into:

$$\alpha[k] = \operatorname{erf}\left(\frac{|y_{OI}[k-1] - y_I[k]|}{\sigma_{Y_I}[k]\sqrt{2}}\right)\quad (13)$$

We also note that, for a stable control system, the gain of the noise component propagated into the system must not exceed the unitary value when reaching the output signal. Otherwise, the noise will be amplified in each complete loop performed, reaching an unstable system. Therefore, the following relation is valid in steady-state closed-loop operation:

$$\frac{y}{n} = -|G|, \quad |G| \leq 1, \quad (14)$$

where G is the plant transfer function.

Thus, the influence of the measurement noise on the measured output can be represented as:

$$y_I = (1 - G)n \quad (15)$$

which in terms of variance can be written as:

$$\sigma_{Y_I}^2 = (1 - G)^2 \sigma_n^2, \quad 0 \leq \sigma_{Y_I}^2 \leq 4\sigma_n^2 \quad (16)$$

Therefore, we arrive at the final expression for the weight coefficient α by rewriting (13) assuming the worst-case scenario to express $\sigma_{Y_I}^2$ in terms of σ_n^2 :

$$\alpha[k] = \operatorname{erf}\left(\frac{|y_{OI}[k-1] - y_I[k]|}{2\sigma_n\sqrt{2}}\right) \quad (17)$$

Remark 2. If there is no native availability of the Gauss error function in the PLC, Eq. (17) can be implemented using the well-known Taylor–Maclaurin series expansion of the Gauss error function [18].

Remark 3. In many practical cases the closed-loop system has already an EMA filter with α_s as the original time constant of the filter. For those cases, the proposed solution to keep the good relation between time response and noise attenuation of the EMA is to adapt (17) considering $\max(\alpha) = \alpha_s$ and keeping the original closed-loop performance for statistically representative disturbances:

$$\alpha[k] = \operatorname{erf}\left(\frac{|y_{OI}[k-1] - y_I[k]|}{2\sigma_n\sqrt{2}}\right) \alpha_s \quad (18)$$

3. Numerical and experimental tests

The main purpose of applying a filter in a control loop is to reduce undesired control activity and its related performance issues. However, filtering techniques require a trade-off between noise reduction and output performance of the controller because part of the desired control action may be lost along with the filtering. In addition, depending on the phase margin of the controlled

system, the application of a first-order filter may deteriorate the control signal by making the controller react to the feedback later than expected. The proposed adaptive filter can relieve this trade-off for less stable systems by maintaining performance while reducing control signal noise.

To illustrate this behavior, in this section the performance of the proposed adaptive filter is first compared to the first-order filter designed following [7,8] and an Extended Kalman Filter (EKF). Numerical simulations were carried out for two illustrative systems in order to evaluate the impact of each filter on both noise attenuation and system output when compared to the unfiltered case. After that, numerical and experimental tests were conducted for a coupled tank system to evaluate its closed-loop performance when compared to other filters. In all cases, a classical parallel PID controller (see [Fig. 3](#)) was employed in the system.

As quantitative criteria for comparison, we considered in the following examples the Integrated Absolute Error (IAE) and the Integrated Squared Error (ISE) to evaluate output performance and the Total Variation (TV) of the control signal for the actuator usage [19,20]. It is also worth noting that for all of these criteria, lower values mean better performance.

3.1. Fourth-order system

Consider the fourth-order transfer function borrowed from [7,8]:

$$G(s) = \frac{1}{(s+1)^4}, \quad (19)$$

for which a PID controller has been designed yielding $K_P = 2.510$, $K_I = 0.554$ and $K_D = 2.830$. The variance of the noise applied to the system output is $\sigma_n^2 = 0.08$.

For comparison purpose, we consider four cases for the same control loop: unfiltered; first-order filter with fixed time constant; Extended Kalman Filter (EKF); and the proposed adaptive filter (1). The fixed time constant of the first-order filter is 0.105 s, obtained by following the procedures presented in [7,8].

The EKF is implemented for comparison with a filter designed in time-domain. Note that, if the dynamics (19) are known and the process and measurement covariances are given, the Kalman filter provides the minimum mean-square error estimator. Thus, for a fair comparison with the proposed model-free adaptive filter, we assume the dynamics are unknown and the EKF is used for both state estimation and system parameter identification following the schema presented in [13, Section 8.3]. For this purpose, an augmented nonlinear model is adopted with the following state space representation [13]

$$\begin{bmatrix} x \\ \theta \end{bmatrix}[k+1] = \begin{bmatrix} A(\theta) & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} x \\ \theta \end{bmatrix}[k] + \begin{bmatrix} B(\theta) \\ 0 \end{bmatrix} u[k] + d[k] \quad (20)$$

$$y_I[k+1] = [C(\theta) \quad 0] \begin{bmatrix} x \\ \theta \end{bmatrix}[k] + n[k] \quad (21)$$

where x is the vector of estimated internal states of the open loop system, $A(\theta)$, $B(\theta)$, and $C(\theta)$ are known matrix-valued functions of the unknown constant vector θ to be estimated. The system disturbances and output AWGN are represented by d and n respectively, the latter with a known variance as required by the adaptive filter algorithm. Since no prior information about the system dynamics is available, we adopted an observable canonical form of third order for $(A(\theta), B(\theta), C(\theta))$.

A step disturbance in the control signal and in the reference is applied at time 5 s and 15 s, respectively. The time-simulation for each case is illustrated in [Fig. 4](#) and the performance indexes IAE, ISE and TV depicted in [Table 1](#). We can see that the proposed adaptive filter yields the highest noise attenuation (lowest TV)

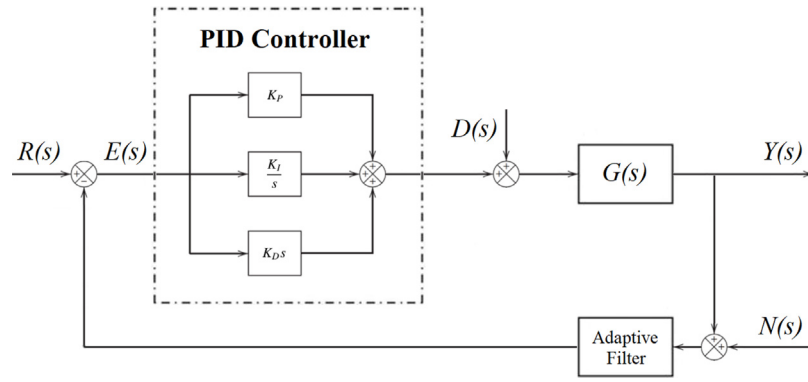


Fig. 3. Block diagram of the classical parallel PID controller.

Table 1

Control performance indexes for the unfiltered case, first-order filter with fixed time constant, EKF and adaptive filter (1) applied in system (19).

	Unfiltered	First-order filter	Extended Kalman filter	Adaptive filter
IAE (Y)	23.1404	24.0057	23.9050	23.3323
ISE (Y)	41.6751	44.5531	43.3173	42.4387
TV (U)	393.4215	222.1850	220.7500	198.0592

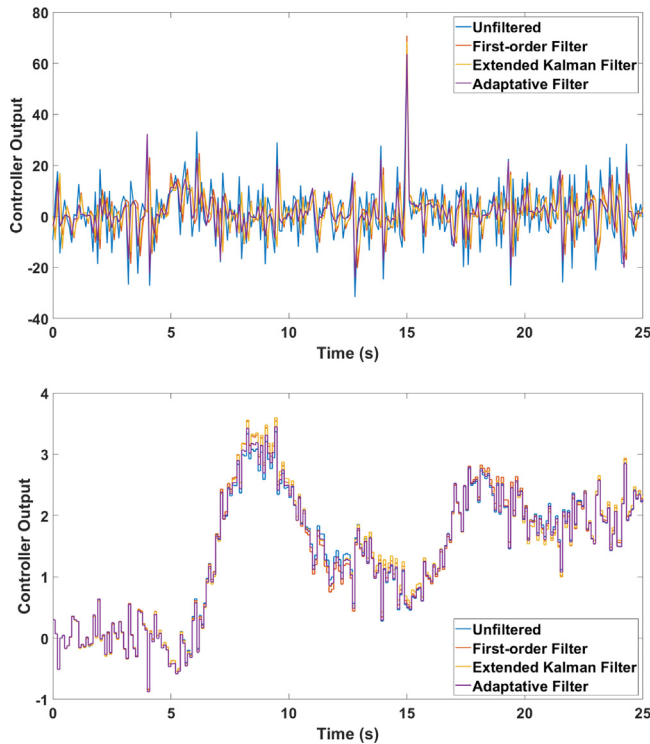


Fig. 4. Control signal (top figure) and measured output (bottom figure) of the closed-loop for the fourth order system (19): unfiltered (blue), fixed time constant first-order filter (red), EKF (yellow) and the adaptive filter (1) (purple). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

with less impact on the system output performance (lowest IAE and ISE). Furthermore, the output signals on Fig. 4 show that the adaptive filter can follow the unfiltered case closer than the other two filters, although each filter was able to maintain output performance similar to the unfiltered case.

The evolution of the adaptive filter parameter $\alpha[k]$ over time is shown in Fig. 5. Compared to the first-order filter, it is clear how the adaptive filter adjusts itself to become more or less

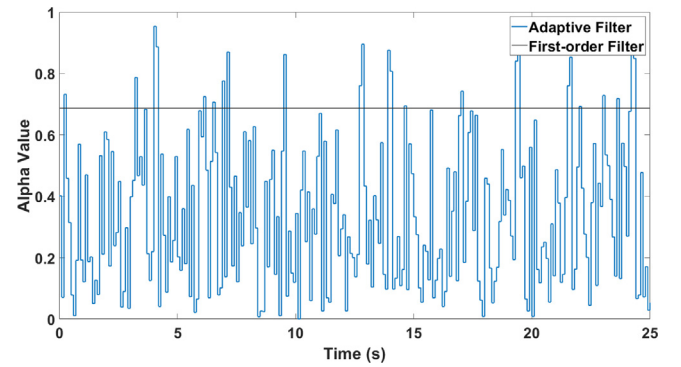


Fig. 5. Adaptive filter parameter $\alpha[k]$ and the time constant of the first-order filter for Example 3.1.

reactive throughout the simulation. In contrast, the first-order filter can only maintain its sensibility to noise. The mean value of α can also be adjusted according to the variance configured for the measurement signal - a higher variance configuration yields lower values for α . The adaptive filter aims to keep this parameter low during stationary state and high during transient state. However, the difference between minimum and maximum values of measurement noise still cause peaks of higher reactivity in the adaptive filter.

3.2. Time-delay system

Consider the system described by a time-delay transfer function as follows [7,8]:

$$G(s) = \frac{1}{s(0.5s + 1)} e^{-0.5s} \quad (22)$$

for which a PID controller has been designed giving $K_P = 0.866$, $K_I = 0.073$ and $K_D = 0.435$.

As a matter of comparison, we also considered the same four cases of the previous example. The fixed time constant of the first-order filter is 0.0753 s, obtained by following the procedures presented in [7,8]. The EKF was designed as expressed in (20)

Table 2

Control performance indexes for the unfiltered case, first-order filter with fixed time constant, EKF and adaptive filter (1) applied in system (22).

	Unfiltered	First-order filter	Extended Kalman filter	Adaptive filter
IAE (Y)	49.0398	49.1254	48.8619	49.1353
ISE (Y)	83.6011	83.6049	82.8670	83.6274
TV (U)	128.6181	64.6499	69.1913	61.9779

and (21), with the same considerations regarding the estimated parameters.

A step disturbance is applied in the control signal and in the input reference at 1 s and 30 s, respectively. In addition, to test the robustness of the proposed filter to non-white noise, the output signal was subject to an additive colored noise obtained from an AWGN signal applied in a first-order transfer function with a time constant of 0.1 s and a gain that keeps the same measurement noise variance applied to the previous system, which is $\sigma_n^2 = 0.08$.

The control signals and the measured outputs of the closed-loop system to a load disturbance and a reference step are illustrated in Fig. 6 for all cases. The control performance indexes are shown in Table 2.

Similarly to the previous example, the evolution of the adaptive filter parameter $\alpha[k]$ over time is shown in Fig. 7. The adaptive filter yields the best noise attenuation (lowest TV) with minimal impacts on the output response (similar IAE and ISE) when compared to the other filters. In this case the EKF performed slightly worse than the first-order filter, and the unmatched dynamics for the time-delayed system brought the worst actuator wear reduction of the three filters (highest TV). Finally, we can still observe that although the first-order filter with a fixed time constant and the EKF can effectively reduce the impact of the noise on the control signal, the proposed adaptive filter can achieve the smoothest control signal with minimal impacts on the output of the system.

3.3. Coupled tank system

Consider the water tank system composed of two tanks aligned vertically connected to each other by an adjustable valve shown in Fig. 8. Water is pumped into the upper tank and then flows through the valve into the lower tank. Finally, its discharged back into the pump reservoir through a small opening located at the bottom of the lower tank. The differential equations for this type of system have been deduced in [21].

By choosing a classical parallel PID controller to evaluate its control performance with and without the proposed adaptive filter, numerical and experimental tests were carried out for the coupled tank system. The trajectory control task consists of adjusting the water level in the lower tank.

The coupled tank system used for validation was obtained within a pilot plant of a quadruple-tank process shown in Fig. 8. Each tank is 68 cm high and has electrical valves to regulate the input flow from the two pumps. The input power to each pump is controlled by two different variable frequency drives. The control loop and the filter were configured on the Programmable Logic Controller from B&R Automation, model X20cCP1584, while MATLAB/Simulink captured the data for later analysis. There is substantial measurement noise from the piezoresistive level sensors located at the bottom of each tank. The variance of the noise in open-loop stationary state of the sensor in the lower tank was determined and used as the configuration parameter σ_n^2 for the adaptive filter. The noise pattern was supposed AWGN and analyzed with both numerical and experimental results. The measurement noise variance considered in both cases is $\sigma_n^2 = 0.4703$.

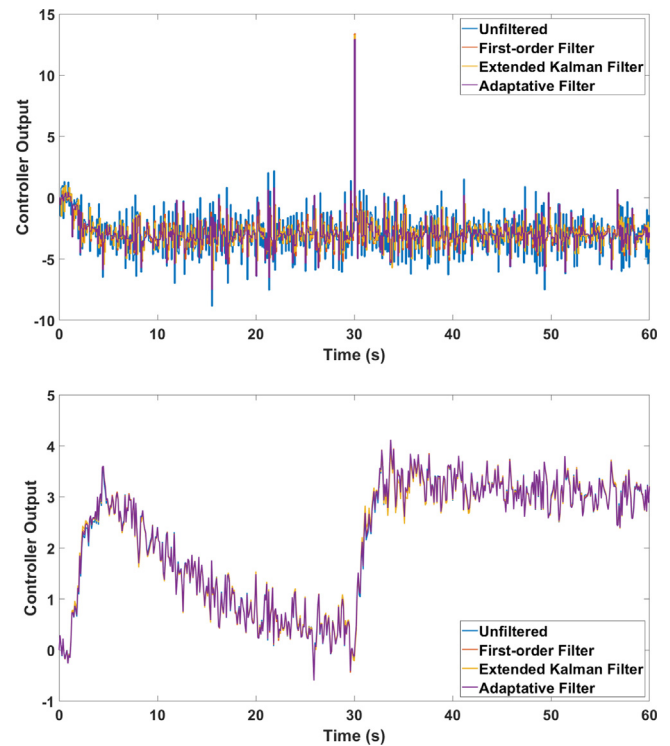


Fig. 6. Control signal (top figure) and measured output (bottom figure) of the closed-loop for the time-delayed system (22): unfiltered (blue), fixed time constant first-order filter (red), EKF (yellow) and the adaptive filter (1) (purple). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

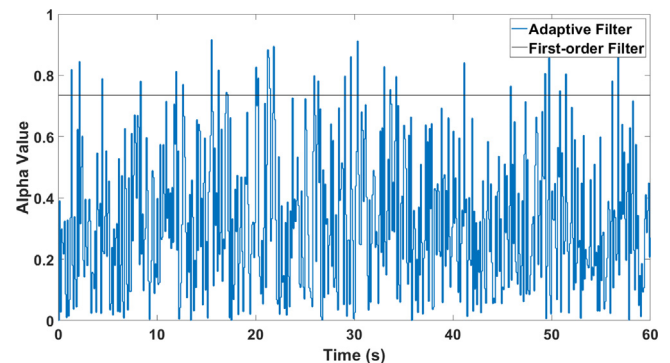


Fig. 7. Adaptive filter parameter $\alpha[k]$ and the time constant of the first-order filter for Example 3.2.

3.3.1. Numerical results

An identification procedure was first carried out by using the MATLAB's System Identification Toolbox to obtain a linearized second-order system around an operating point for the level control of the lower tank. The identified transfer function is given

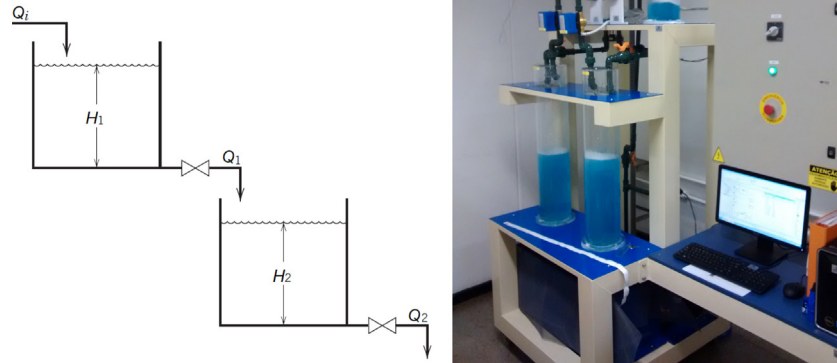


Fig. 8. Schematic of the two-tank system (left) and experimental bench for the filter implementation (right).

Table 3
Control performance indexes (Numerical simulation).

	Unfiltered	First-order filter	Extended Kalman filter	Adaptive filter
IAE (Y)	7.0091×10^3	7.0095×10^3	6.9868×10^3	7.0115×10^3
ISE (Y)	3.7666×10^4	3.7698×10^4	3.7157×10^4	3.8138×10^4
TV (U)	7.2685×10^3	6.9828×10^3	7.2036×10^3	3.6559×10^3

by:

$$G(s) = \frac{0.002677}{(s + 0.0565)(s + 0.0192)} \quad (23)$$

The PID controller gains were tuned using the Internal Model Control (IMC) approach described in [22], yielding $K_P = 0.3105$, $K_I = 0.0044$ and $K_D = 4.1019$. Fig. 9 shows the control signal and output response to load disturbances for each of the four proposed cases applied in the previous examples.

As expected, the closed-loop output responses are numerically similar in all cases, while the effects of measurement noise on the dynamic behavior of the actuator are better mitigated by using the proposed adaptive filter.

To evaluate the results quantitatively, the performance indexes IAE, ISE and TV are presented in Table 3. Note that the total variation of the control signal has been reduced to approximately half of the unfiltered case by using the proposed adaptive filter, corresponding to the findings in previous numerical simulations. The first-order filter yielded the second best indexes regarding noise reduction. The EKF could not achieve better results because of the model differences between the estimated parameters and the configured dynamics. Despite the lower noise in the control signal, the output performance indexes confirm that all filters were able to maintain the unfiltered dynamics.

3.3.2. Experimental results

The PID controller was experimentally implemented and tested unfiltered. Each of the filters was implemented with the same control gains used in the numerical simulations on the experimental bench. Each experiment was conducted successively to minimize the effects of external disturbances on the system. The same input references were applied to the real system according to the numerical simulation.

The experimental results, illustrated in Fig. 10, agree with the conclusions obtained in the numerical simulation. More precisely, the use of the proposed adaptive filter reduced the effects of the measurement noise on the dynamic behavior of the actuator

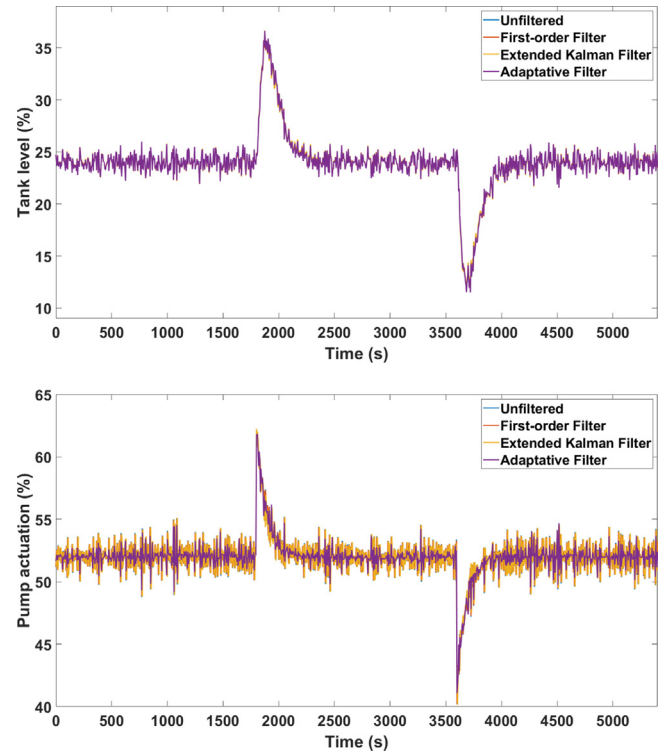


Fig. 9. Time response of the closed-loop output (top figure) and the control signal (bottom figure) to a load disturbance step in the instants 1800 s (amplitude 10) and 3600 s (amplitude -10) for each of the proposed cases in the numerical simulation.

without changing the closed-loop output response significantly. The performance indexes are available in Table 4 as usual. We

Table 4

Control performance indexes (Experimental test).

	Unfiltered	First-order filter	Extended Kalman filter	Adaptive filter
IAE (Y)	7.9862×10^3	7.5603×10^3	7.6932×10^3	8.4728×10^3
ISE (Y)	3.8205×10^4	3.7532×10^4	3.8245×10^4	3.8965×10^4
TV (U)	1.5442×10^3	1.4585×10^3	1.4039×10^3	1.0616×10^3

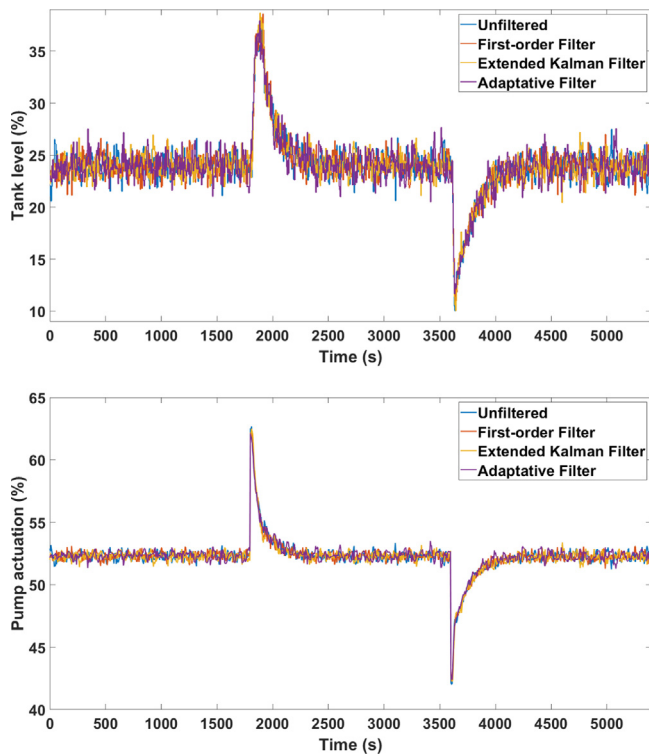


Fig. 10. Time response of the closed-loop output (top figure) and the control signal (bottom figure) to a load disturbance step in the instants 1800 s (amplitude 10) and 3600 s (amplitude -10) for each of the proposed cases in the experimental test.

observe that the performance indexes presented for the adaptive filter in Tables 1–3 for the numerically simulated cases show the reduction of the TV index to roughly half of the unfiltered case. A reduction of 31,2% on the TV index was observed for the experiment.

4. Industrial application

This section presents the application of the filter in four control loops of an industrial plant. The filter algorithm has been implemented in the supervisory system to analyze the effects of reducing the control signal variance and, consequently, actuator wear. The adaptive filter performance is compared to the original design for each of the control loops. Finally, the standard deviation σ_n for each loop, used in adaptation law, was obtained from the output measurements and set according to individual specifications.

4.1. Industrial plant

The industrial plant is an atmospheric tower from a distillation unit belonging to a Brazilian oil refinery (see Fig. 11) with an average throughput of 30,000 m³/d (188,700 bpd) of oil. Diesel is the main refinery product (50% of production), followed by gasoline (25%). The remainder production is divided into jet



Fig. 11. Distillation unit tower.

fuel (kerosene), fuel oil, liquefied petroleum gas, asphalt, petrochemical naphtha and propylene. The distillation unit started its operation in 1993, being able to process 125,000 bpd of crude. Its current capacity is 132,000 bpd and feed switches occur 3 to 4 times per week. The unit has six product draws (Liquified Petroleum Gas, Light Naphtha, Heavy Naphtha, Kerosene, Light Diesel and Heavy Diesel) plus the atmospheric residue which is sent to a Residue Fluid Catalytic Cracking Unit. The simplified flow diagram of the plant is shown in Fig. 12.

The distillation unit variables are controlled by a commercial DCS from Emerson Electric Co. (Delta V Emerson), which engineering station operates in a Windows environment, making it easy to enter non-native logic and controls. The adaptive filter was programmed directly in each DCS PID control module. The filter parameter $\alpha[k]$ was converted to a first-order time constant and sent to the corresponding native PID port, as illustrated in Fig. 13 at the bottom right, following an appropriated block execution order for each time scan.

4.2. Industrial results

The adaptive filter was implemented in four control flow loops of the atmospheric distillation tower. The loops are not detailed due to confidentiality reasons and are referenced by Loops 1 to 4. In the time responses, the output of the closed-loop system is denoted by PV, the set-point by SP, and the control signal by OP. The adaptive filter requires a configuration for the expected standard deviation of the measurement noise present in the system output. This parameter directly affects the efficiency of

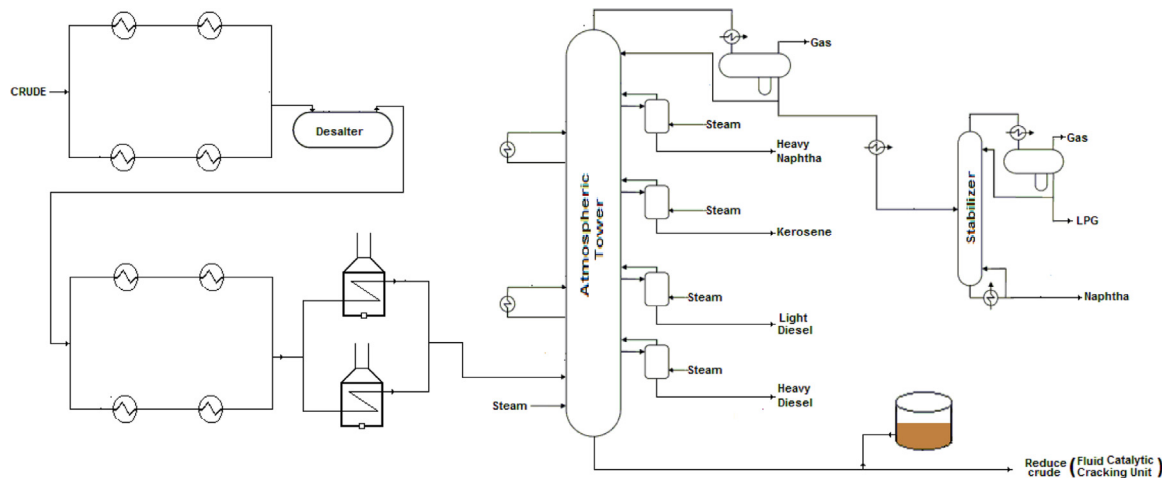


Fig. 12. Distillation unit simplified scheme.

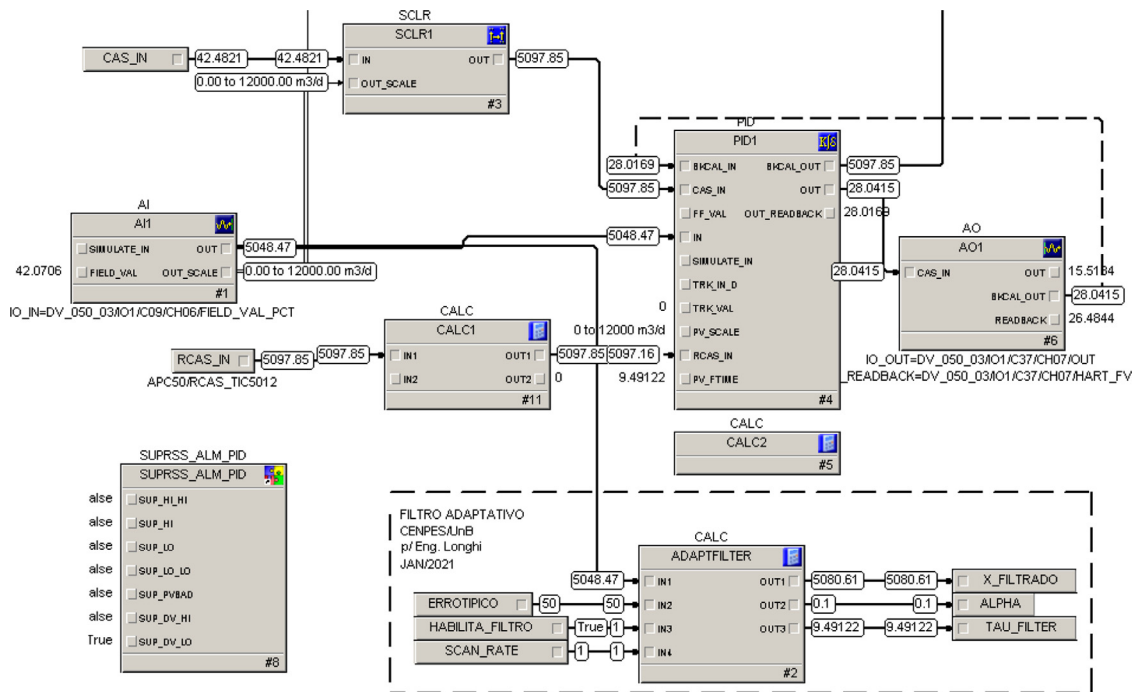


Fig. 13. Detail of adaptive filter implementation in a control loop DCS module.

the filter in regards to noise attenuation while preserving output performance.

The filter parameter σ_n of Loop 1 was set higher than the observed output standard deviation, leading to modifications in the controller's closed-loop response. Although maintaining the original controller response is one of the main advantages of using this filter, it is also possible to adjust its behavior and reach much higher noise reductions. Therefore, the value set for σ_n is essential to achieve the best compromise between noise attenuation and control loop response for each specific case.

Table 5 shows the performance indexes of Loops 1, 2, 3 and 4 for the closed-loop with the original PID controller (unfiltered) and with the adaptive filter. Observe that using the adaptive filter in Loop 1 yielded 70% of noise reduction compared to the initial setup. This reduction is higher than the ones observed in the other loops due to the choice of σ_n higher than the observed value. However, we also note that the IAE performance index of the output is also affected. This effect is due to the tracking problem,

as we can see in Fig. 14 that shows a good follow-up of the PV to the SP.

The standard deviation adjusted for the filters in Loop 2 and 3 were close to the expected standard deviation measured in their sensors, as was done in experimental tests of Section 3. This procedure ensures the control loop response is maintained while the filter achieves some attenuation of the control signal. This configuration is most appropriate when keeping the current output response is strictly necessary. As a result, we observe in Figs. 15 and 16 a slight reduction in the TV of the control signal with little differences in the system response. Loop 4 is an example of a case where the filter shows minimal effects in noise attenuation. In this control loop, a sticktion compensator constantly modifies the PV avoiding the stationary state and preventing the actuator from sticking, as noticed in the OP signal of Fig. 17. It is still possible to obtain minor noise attenuation, but the premise of keeping the original controller response is of utmost importance for this unusual case.

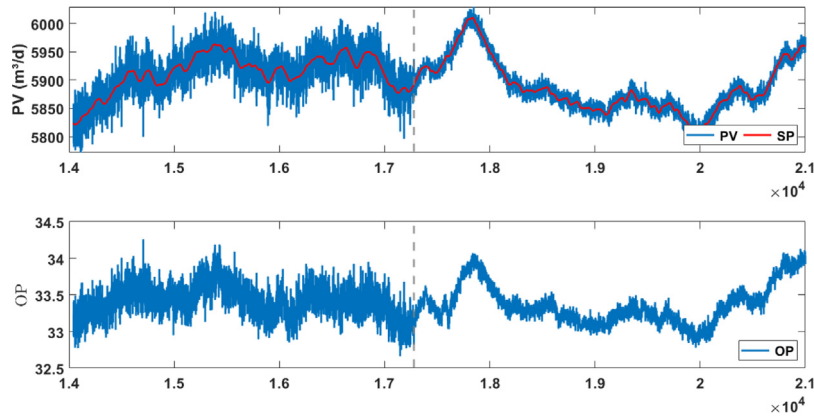


Fig. 14. The top figure shows the time evolution of the output (PV) and set-point (SP) signals in m^3/d , and the bottom figure shows the control signal (OP) in %. The dashed vertical line delimits the time evolution before (left) and after (right) the implementation of the adaptive filter in Loop 1. The time scale is given in seconds.

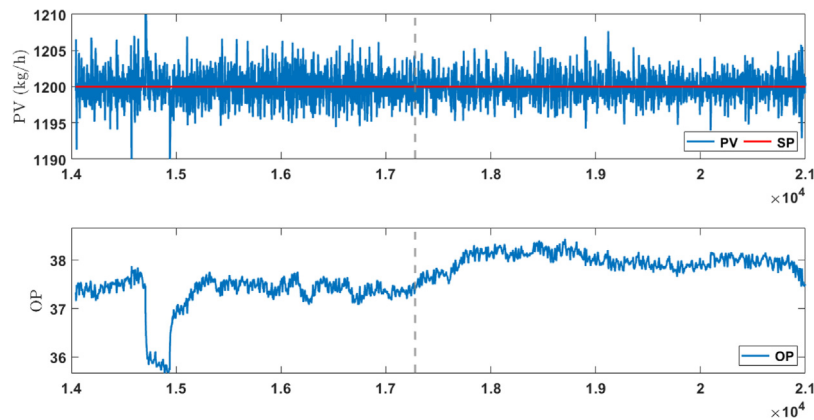


Fig. 15. The top figure shows the time evolution of the output (PV) and set-point (SP) signals in kg/h , and the bottom figure shows the control signal (OP) in %. The dashed vertical line delimits the time evolution before (left) and after (right) the implementation of the adaptive filter in Loop 2. The time scale is given in seconds.

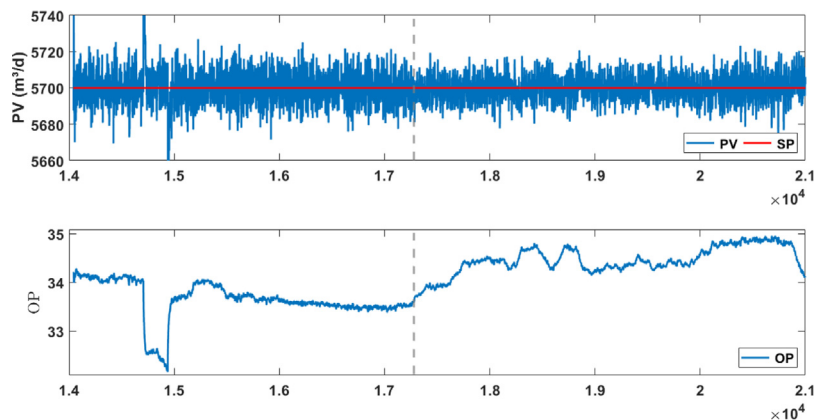


Fig. 16. The top figure shows the time evolution of the output (PV) and set-point (SP) signals in m^3/d , and the bottom figure shows the control signal (OP) in %. The dashed vertical line delimits the time evolution before (left) and after (right) the implementation of the adaptive filter in Loop 3. The time scale is given in seconds.

5. Conclusions

In this paper, a model-free statistical-based adaptive filter is proposed to reduce actuator wear while simultaneously guaranteeing the performance of the closed-loop system. The adaptation

law of the filter parameter is model-free and depends on the standard deviation of the measurement noise, which can be obtained straightforward. Therefore, the main highlights of the proposed approach are the industrial scalability, a straightforward implementation on most of the Programmable Logic Controllers, and an

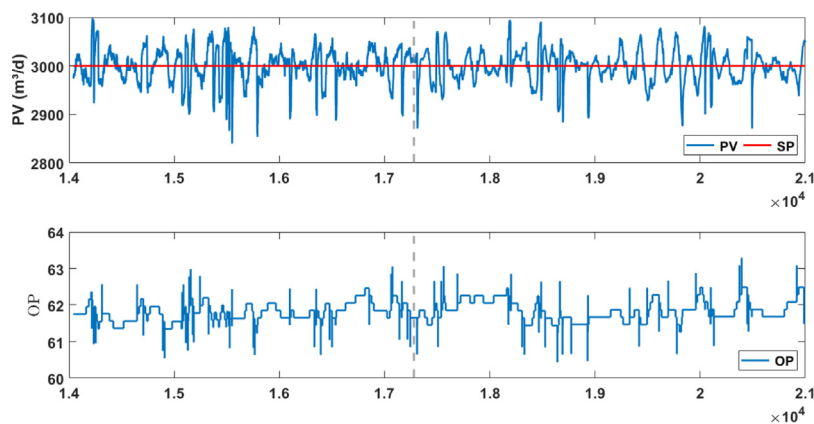


Fig. 17. The top figure shows the time evolution of the output (PV) and set-point (SP) signals in kg/h, and the bottom figure shows the control signal (OP) in %. The dashed vertical line delimits the time evolution before (left) and after (right) the implementation of the adaptive filter in Loop 4. The time scale is given in seconds.

Table 5

Control performance indexes of Loops 1, 2, 3 and 4 for the closed-loop with the original PID controller (unfiltered) and with the adaptive filter.

		Unfiltered	Adaptive filter
Loop 1	IAE (Y)	8.3377×10^5	3.1774×10^5
	ISE (Y)	2.5508×10^7	0.3676×10^7
	TVC (U)	8.1148×10^3	2.4682×10^3
Loop 2	IAE (Y)	6.9406×10^4	6.2096×10^4
	ISE (Y)	1.9213×10^5	1.4243×10^5
	TVC (U)	8.6647×10^2	7.1323×10^2
Loop 3	IAE (Y)	2.9155×10^5	2.5208×10^5
	ISE (Y)	3.4710×10^6	2.4132×10^6
	TVC (U)	6.0787×10^2	4.6075×10^2
Loop 4	IAE (Y)	1.2849×10^6	1.2264×10^6
	ISE (Y)	6.0891×10^7	5.1078×10^7
	TVC (U)	1.1115×10^3	1.0040×10^3

intuitive tuning since the only parameter to set is the measurement noise standard deviation of the sensor. Compared to other approaches in the literature, the main points are the increased performance and the extreme simplicity for practical use since it is a model-free technique.

As the PID controller is the most widely used control strategy in industry, the performance of the proposed filter has been evaluated through numerical simulations, experimental tests, and practical use in an oil refinery. The adaptive filter was efficiently implemented to lengthen the life of the actuator while ensuring the same system performance as the original closed-loop, even in the presence of a stiction compensator. Further analysis will include long-term results of the methodology in an industrial plant to quantify the financial impact in the maintenance costs of control equipment and implementation in other process plants. Future works will include statistical-based filtering strategies for fault detection in dynamic process, and the development of adaptation laws better suited to adjust the $\alpha[k]$ parameter.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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